Math 107 Midterm 1 practice

Last updated: September 24, 2026.

Based on Lectures 01–14 and Quizzes 1–4. Use with the study guide. Exam: Friday, October 2 · in class (not posted) · day page.

Three types of problems will appear on Midterm 1: short answer definitions and meaning, computational and applied problems.

A. Definitions and meaning

Write a short precise answer (one or two sentences, or a formula). No long essays.

A1. What is an n-vector? Give one non-math example from class (with units on the entries).

A2. What is a **linear combination** of vectors $a_1, \dots, a_k$?

A3. Define the **inner product** (dot product) of two n-vectors. When are two vectors **orthogonal**?

A4. When is a function $f : \mathbb{R}^n \to \mathbb{R}$ called **linear**? How does **affine** differ?

A5. Define the **Euclidean norm** $\|x\|$ and the **distance** between x and y.

A6. Write the formula for **cosine similarity** of nonzero a and b. What range of angles $\theta = \angle(a, b)$ do we use?

A7. What is a **(distance) nearest neighbor** of a query x among candidates $z_1, \dots, z_m$?

A8. In one sentence: what is the goal of **clustering** N vectors into k groups?

A9. State the definition of **linear dependence** of $a_1, \dots, a_k$. What does **independent** mean?

A10. Define $\text{span}\{a_1, \dots, a_k\}$.

A11. What is a **subspace** of $\mathbb{R}^n$? Why is the line $y = 2x + 1$ in $\mathbb{R}^2$ **not** a subspace?

A12. What is a **basis** of a subspace V? What is $\dim V$?

A13. When is a list $q_1, \dots, q_k$ **orthonormal**? If it is an orthonormal basis of a subspace containing v, how do you find the coefficients of v?

A14. True or false, with a one-line reason: "If $\|a - b\|$ is small, then $\angle(a, b)$ must be near 0."

A15. True or false, with a one-line reason: "Every spanning list for V is a basis of V."

A16. *k-means vocabulary.*

(a) In k-means, what does the integer k stand for?
(b) After vectors have been assigned to k clusters, what is the formula for the **center (centroid)** of a nonempty cluster?
(c) Name the **two steps** that k-means alternates after an initial assignment (or after choosing initial centers).

A17. For each expression, say whether the result is a **scalar**, an **n-vector**, a **set of vectors**, or **undefined** (brief reason if undefined). Assume a, b, u, v are n-vectors and α is a scalar, unless noted.

(a) $a^\mathsf{T} b$ (dot product of two n-vectors)

(b) $\|a\|$

(c) $a + b$

(d) αa

(e) cosine similarity of nonzero a and b

(f) $\angle(a, b)$

(g) $\text{span}\{a, b\}$

(h) $a^\mathsf{T} b + c$ when c is an n-vector

(i) $a + b^\mathsf{T}$ (treating b^T as a row)

(j) $\mathrm{rms}(a)$

(k) a k-means centroid of a nonempty cluster of n-vectors

(l) $\dim \mathrm{span}\{a, b\}$

(m) $a^\mathsf{T} b$ when a is a 2-vector and b is a 3-vector

(n) unit vector $\hat{a} = a/\|a\|$ when $a \neq 0$

B. Computational

Show your work for each calcualtion.

Vectors, combinations, dot products

B1. Let $u = (3, -1, 2)$ and $v = (0, 4, -1)$. Compute $u + v$, $2u - v$, and $u^\mathsf{T} v$.

B2. Write $w = (5, 1)$ as a linear combination of $a = (1, 0)$ and $b = (1, 1)$.

B3. Are $(1, 2)$ and $(-2, -4)$ orthogonal? Compute the dot product.

B4. Let $x = (2, 1, -1)$. Compute $\|x\|$ exactly (simplified radical OK).

B5. Distance between $p = (1, 1)$ and $q = (4, -3)$.

B6. Cosine of the angle between $a = (1, 0)$ and $b = (1, 1)$. Angle in degrees?

B7. Unit vector in the direction of $(3, 4)$.

B8. $x = (1, -1, 1, -1)$. Compute $\mathrm{avg}(x)$ and $\mathrm{rms}(x)$.

B9. Is $f(x) = 3x_1 - x_2 + 2$ linear or affine? Brief reason.

B10. Is $g(x) = x_1 x_2$ linear? Test with one superposition counterexample or explain.

B11. Score $s = w^\mathsf{T} r$ with $w = (0.5, 0.5)$ and $r = (80, 90)$. What is s?

B12. Boolean vectors $x = (1, 0, 1, 1)$ and $y = (1, 1, 0, 1)$. What is $\|x - y\|$?

Nearest neighbors, scale, angle vs distance

B13. Query $x = (1, 2)$; candidates $z_1 = (0, 2)$, $z_2 = (3, 2)$, $z_3 = (1, 3)$. Which is nearest in Euclidean distance? Give the three distances.

B14. Same x and candidates as B13, but replace every first coordinate t by $10t$. Which is nearest now? Did it change?

B15. $x = (1, 0)$, $z_1 = (2, 0)$, $z_2 = (0.5, 0.5)$. Distance-NN and angle-NN of x? Same or not?

B16. Cosine similarity of $(2, 1)$ and $(4, 2)$. What does that tell you about the angle?

k-means

B17. Points $x_1 = (0,0)$, $x_2 = (0,2)$, $x_3 = (6,0)$, $x_4 = (6,2)$; representatives $z_1 = (1,1)$, $z_2 = (5,1)$. Assign each point to the nearest z_j.

B18. Using your clusters from B17, update each z_j to the mean of its points.

B19. After B18, do one more assignment step. Does any point switch clusters?

B20. A cluster of 2-vectors has points $(2,0)$ and $(4,2)$. What is the centroid?

B31. *k-means worksheet (2-vectors).* Five vectors are initially assigned to $k = 2$ clusters as in the table. Compute the centroids, fill the distance table (Euclidean), then revise each cluster ID. Some entries are already filled — complete the blanks.

Vector	Initial Cluster ID	Revised Cluster ID
$x_1 = (0,1)$	1	
$x_2 = (2,1)$	1	
$x_3 = (8,1)$	2	2
$x_4 = (9,1)$	2	
$x_5 = (-2,1)$	1	

Center of initial cluster 1: $c_1 =$ _____

Center of initial cluster 2: $c_2 = (8.5,\ 1)$ (given)

Distances	x_1	x_2	x_3	x_4	x_5
c_1		2			
c_2	8.5		0.5		10.5

(Exact values or one decimal place are fine.)

B32. *k-means worksheet (3-vectors).* Four vectors in $\mathbb{R}^3$, $k = 2$. Compute both centroids from the initial assignment, fill **all** distances, then revise the cluster IDs.

Vector	Initial Cluster ID	Revised Cluster ID
$x_1 = (0,0,0)$	1	
$x_2 = (6,0,0)$	1	
$x_3 = (1,0,0)$	2	
$x_4 = (0,8,0)$	2	

$c_1 =$ _____
$c_2 =$ _____

Distances	x_1	x_2	x_3	x_4
c_1				
c_2				

Dependence, span, basis

B21. Are $(1,0)$ and $(0,1)$ independent in $\mathbb{R}^2$? Brief reason.

B22. Are $(2,4)$ and $(1,2)$ independent? If dependent, give a dependence relation.

B23. Find scalars not all zero with $\beta_1(1,1) + \beta_2(1,-1) + \beta_3(0,2) = 0$.

B24. Is $(3,3)$ in $\text{span}\{(1,1)\}$? If yes, give the coefficient.

B25. Is $(1,0)$ in $\text{span}\{(1,1),(1,-1)\}$? Coefficients or reason.

B26. Is $(1,0,0)$ in $\text{span}\{(1,1,0),(0,1,1)\}$? Reason.

B27. Give a basis for $\text{span}\{(1,2,0),(2,4,0),(0,0,1)\}$ and state the dimension.

B28. Is $\{(1,0),(0,1),(1,1)\}$ a basis of $\mathbb{R}^2$? Why or why not?

B29. Are $q_1 = (1,0)$ and $q_2 = (0,1)$ orthonormal? For $v = (3,-2)$, find $\beta_1 = q_1^{\mathsf{T}} v$ and $\beta_2 = q_2^{\mathsf{T}} v$.

B30. Are $(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$ and $(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}})$ orthonormal?

C. Applied

Show reasoning. These mix ideas or need a careful choice of tool.

C1. Let $p = (1,1)$, $q = (10,10)$, $r = (1,-1)$.

(a) Which of q, r is closer to p in **Euclidean distance**?
(b) Which is more similar to p in **direction**?
(c) In one sentence: when would you prefer angle over distance for "similarity"?

C2. A café stores each customer as (coffee count, pastry count). Someone adds a third feature "total items" = coffee + pastry.

(a) Are the three feature directions dependent? Explain.
(b) Why might this hurt nearest-neighbor recommendations that use Euclidean distance?

C3. Data: $x_1 = (0,0)$, $x_2 = (1,0)$, $x_3 = (0,1)$, $x_4 = (10,10)$. Run one k-means **assignment** with $k = 2$ and initial $z_1 = (0,0)$, $z_2 = (9,9)$. Which points go to which cluster? What goes wrong conceptually if you wanted two "local" clusters near the origin and ignored the outlier's pull on a centroid later?

C4. Let $a = (1,2)$, $b = (2,1)$, $c = (3,3)$.

(a) Show $\{a,b,c\}$ is dependent by exhibiting a dependence relation.
(b) Show $\{a,b\}$ is a basis of $\mathbb{R}^2$.
(c) Express c in that basis (coefficients).

C5. Decide which are subspaces of $\mathbb{R}^2$: (i) $\text{span}\{(2,1)\}$; (ii) all (x,y) with $y = 3x + 1$; (iii) $\{(0,0)\}$; (iv) the unit circle. One sentence each.

C6. Feature vectors for documents (word counts) can have very different total lengths. Why is **cosine similarity** often preferred to Euclidean distance for "similar documents"?

C7. $f(x) = \|x - c\|^2 - \|x - d\|^2$ for fixed c, d. Expand and show f is **affine** in x (find a and b in $a^{\mathsf{T}}x + b$).

C8. You normalize every candidate to unit length, then pick the Euclidean nearest neighbor to a query x that is already a unit vector. What comparison are you effectively making?

C9. Let $V = \text{span}\{(1,0,1),(0,1,1)\}$.

(a) Find $\dim V$.
(b) Give one vector in $\mathbb{R}^3$ **not** in V, with a short reason.

(c) Is $(1, 1, 2)$ in V? If yes, give coefficients.

C10. Mixed "which tool?" — for each task, name the best primary tool from: linear combination, dot product, norm/distance, angle, nearest neighbor, k-means, span, independence, basis.

Task	Tool
Weighted course score from a grade vector	
"Is this third sensor redundant?"	
"Which past day is most like today?"	
"Group songs into two mood types"	
"Can this vector be built from a and b?"	
"Minimal independent spanning set for a plane through the origin"	

C11. *Graph — span.* In $\mathbb{R}^2$, sketch axes and draw $a = (2, 1)$ and $b = (-1, 2)$ as arrows from the origin.

(a) Sketch the line $\text{span}\{a\}$ (the whole line through a, not just the arrow).
(b) Shade or describe $\text{span}\{a, b\}$. Is it all of $\mathbb{R}^2$? Why?
(c) Mark a point v that is in $\text{span}\{a\}$ and a point w that is **not** in $\text{span}\{a\}$.

C12. *Graph — unit vectors and angle.* Let $u = (3, 0)$ and $v = (0, 4)$.

(a) Sketch u, v, and the unit vectors $\hat{u} = u/\|u\|$ and $\hat{v} = v/\|v\|$ from the origin (same axes).
(b) What are $\|\hat{u}\|$ and $\|\hat{v}\|$? What is $\angle(\hat{u}, \hat{v})$?
(c) On your sketch, which looks "closer" to $\hat{u}$: the tip of $\hat{v}$, or the tip of v? Which comparison is Euclidean distance making? Which is angle making?

C13. *Graph — closeness vs direction.* Let $p = (1, 1)$, $q = (3, 3)$, and $r = (1, 0)$.

(a) Sketch p, q, and r from the origin.
(b) By eye (then check with a short computation): which of q, r is closer to p in **distance**? Which is closer in **angle/direction**?
(c) Draw the unit circle lightly and mark $\hat{p}$, $\hat{q}$, $\hat{r}$. Explain in one sentence how the unit circle helps you see directional similarity.

Answers

Check only after a closed-book attempt. Update your definitions/formulas list for every miss.

A — Definitions (key points)

A1. List of n numbers (entries) with meaning/units; e.g. temperatures by hour, or (lat, lon).

A2. $\beta_1 a_1 + \cdots + \beta_k a_k$ for scalars β_i.

A3. $a^\mathsf{T} b = \sum_i a_i b_i$; orthogonal iff $a^\mathsf{T} b = 0$.

A4. Linear: $f(\alpha x + \beta y) = \alpha f(x) + \beta f(y)$ (equivalently $f(x) = a^\mathsf{T} x$). Affine: $a^\mathsf{T} x + b$ with possible nonzero b.

A5. $\|x\| = \sqrt{x^\mathsf{T} x}$; distance $\|x - y\|$.

A6. $a^\mathsf{T} b / (\|a\|\|b\|)$; $\theta \in [0, \pi]$.

A7. Any z_j minimizing $\|x - z_j\|$.

A8. Partition so same-group vectors are close (small within-group distances), with representatives.

A9. Dependent: nontrivial β with $\sum \beta_i a_i = 0$. Independent: only $\beta = 0$ works.

A10. All linear combinations of $a_1, \ldots, a_k$.

A11. Nonempty set closed under linear combinations (contains 0). Line $y = 2x + 1$ misses 0 / not closed under scaling.

A12. Independent spanning list; dimension = number of vectors in any basis.

A13. Unit and pairwise orthogonal; $\beta_i = q_i^\mathsf{T} v$.

A14. **False** — e.g. near-origin opposite directions.

A15. **False** — spanning lists can be dependent.

A16. (a) The number of clusters (groups). (b) The mean of the vectors in that cluster: if cluster j contains $x_{i_1}, \ldots, x_{i_m}$, then $c_j = \frac{1}{m}(x_{i_1} + \cdots + x_{i_m})$. (c) **Assignment** (each point to the nearest current center) and **update** (replace each center by the mean of its cluster).

A17. (a) **scalar** (b) **scalar** (c) n**-vector** (d) n**-vector** (e) **scalar** (f) **scalar** (an angle, in radians or degrees) (g) **set of vectors** (a subspace / span) (h) **undefined** — scalar plus vector (i) **undefined** (sizes/shapes don't match for vector + row; not a valid n-vector sum in this course) (j) **scalar** (k) n**-vector** (l) **scalar** (a nonnegative integer) (m) **undefined** — lengths must match for a dot product (n) n**-vector**

B — Easy computations

B1. $(3, 3, 1)$; $(6, -6, 5)$; -6.

B2. $w = 4a + 1b$ (or $w = 4(1, 0) + (1, 1)$).

B3. Dot $-2 - 8 = -10 \neq 0$; not orthogonal. (They are parallel / dependent.)

B4. $\sqrt{4 + 1 + 1} = \sqrt{6}$.

B5. $\|(-3, 4)\| = 5$.

B6. $1/\sqrt{2}$; 45°.

B7. $(3/5, 4/5)$.

B8. avg $= 0$; rms $= 1$.

B9. **Affine** (nonzero constant term).

B10. **Not linear** (product); e.g. $g(e_1) = 0 = g(e_2)$ but $g(e_1 + e_2) = 1$.

B11. 85.

B12. Disagree in 2 positions; $\|x - y\| = \sqrt{2}$.

B13. Distances 1, 2, 1; tie between z_1 and z_3.

B14. After $\times 10$ on the first feature: distances 10, 20, 1; nearest is z_3 only. **Yes**, changed.

B15. Distance-NN z_2; angle-NN z_1; **not** the same.

B16. Cosine $= 1$; angle 0° (same direction).

B17. Cluster 1: $\{x_1, x_2\}$; cluster 2: $\{x_3, x_4\}$.

B18. $z_1 = (0, 1)$, $z_2 = (6, 1)$.

B19. Same assignment; no switches.

B20. $(3, 1)$.

B31. Initial cluster 1 is $\{x_1, x_2, x_5\}$, so

$$c_1 = \frac{(0, 1) + (2, 1) + (-2, 1)}{3} = (0, 1).$$

Distances:

	x_1	x_2	x_3	x_4	x_5
c_1	0	2	8	9	2
c_2	8.5	6.5	0.5	0.5	10.5

Revised IDs: 1, 1, 2, 2, 1 (no switches).

B32. $c_1 = \frac{1}{2}((0,0,0)+(6,0,0)) = (3,0,0)$;
$c_2 = \frac{1}{2}((1,0,0)+(0,8,0)) = (0.5,\ 4,\ 0)$.

	x_1	x_2	x_3	x_4
c_1	3	3	2	$\sqrt{9+64} = \sqrt{73} \approx 8.5$
c_2	$\sqrt{0.25+16} = \sqrt{16.25} \approx 4.0$	$\sqrt{30.25+16} = \sqrt{46.25} \approx 6.8$	$\sqrt{0.25+16} \approx 4.0$	$\sqrt{0.25+16} \approx 4.0$

Revised: $x_1 \to 1$, $x_2 \to 1$, $x_3 \to 1$ (switches from 2), $x_4 \to 2$.

B21. Yes — standard basis; only the trivial combination gives 0.

B22. Dependent: e.g. $(2,4) - 2(1,2) = 0$.

B23. e.g. $\beta = (1,-1,1)$ because $(1,1) - (1,-1) = (0,2)$.

B24. Yes: $3 \cdot (1,1)$.

B25. Yes: $\frac{1}{2}(1,1) + \frac{1}{2}(1,-1)$.

B26. No (inconsistent third coordinate).

B27. e.g. $\{(1,2,0),(0,0,1)\}$; dim = 2.

B28. No — dependent (third = sum of first two), though it spans $\mathbb{R}^2$.

B29. Yes; $\beta_1 = 3$, $\beta_2 = -2$.

B30. Yes.

C — Harder

C1. (a) r. (b) q. (c) When overall scale should not matter.

C2. (a) Yes — third feature = first + second. (b) Double-counts information and can distort Euclidean distance.

C3. $x_1, x_2, x_3 \to z_1$; $x_4 \to z_2$. Clear discussion of outliers / initialization / choice of k is fine.

C4. (a) $a + b - c = 0$. (b) Two independent vectors in $\mathbb{R}^2$. (c) $c = a + b$.

C5. (i) yes; (ii) no; (iii) yes; (iv) no.

C6. Document lengths vary; cosine compares direction / word pattern.

C7. $f(x) = 2(d-c)^{\mathsf{T}}x + \|c\|^2 - \|d\|^2$.

C8. Angle nearest-neighbor among those candidates.

C9. (a) 2. (b) e.g. $(1,-1,0)$ or $(0,0,1)$. (c) Yes: $(1,0,1) + (0,1,1)$.

C10. Dot product; independence/dependence; nearest neighbor; k-means; span; basis.

C11. (a) Line through the origin along $(2,1)$. (b) All of $\mathbb{R}^2$ — a, b are independent (not parallel). (c) e.g. $v = (4,2)$ on the line; $w = (0,1)$ off the line.

C12. (a) $\hat{u} = (1,0)$, $\hat{v} = (0,1)$. (b) Both norms 1; angle 90°. (c) Tip of $\hat{v}$ is distance $\sqrt{2}$ from tip of $\hat{u}$; tip of v is farther from tip of $\hat{u}$. Distance compares positions of tips; angle compares directions from the origin (here both pairs of directions are orthogonal).

C13. (a) Sketch. (b) Distance: r is closer ($\|p-r\| = 1$ vs $\|p-q\| = 2\sqrt{2}$). Direction: q (same ray as p; angle 0). (c) On the unit circle, $\hat{p} = \hat{q}$ coincide; $\hat{r}$ is elsewhere — directional closeness is closeness of the unit tips.